

Addressed to: Dr. Priya Kulkarni (GP)
Hospital Number: AML002
Date of Letter: 12.03.2024

Dear Dr. Kulkarni,

We are writing to summarize the completed treatment course and current status of **Mr. William Moore (*09.12.1959)**, a 64-year-old gentleman followed in our service for acute myeloid leukemia.

Diagnosis and Risk Stratification

- **Date of diagnosis:** 30.06.2021 after evaluation for fatigue, gingival bleeding, and leukocytosis.
- **Morphology/Flow:** Consistent with AML; flow cytometry positive for CD13, CD33, CD117, HLA-DR, and partial CD64.
- **Cytogenetics:** t(9;11)(p21.3;q23.3) with KMT2A (MLL) rearrangement.
- **ELN 2022 risk:** Intermediate.
- **Molecular panel:** No FLT3-ITD/TKD, no NPM1 or IDH mutations.

Initial Presentation

At diagnosis: WBC $32 \times 10^9/L$ (blasts 52%), Hgb 8.9 g/dL, platelets $41 \times 10^9/L$. LDH 612 U/L. Coagulation normal. Echocardiogram LVEF 58%. CT chest/abdomen without organomegaly.

Treatment Summary

- **Induction (1L):** Standard “7+3” (cytarabine continuous infusion $\times 7$ days, daunorubicin $\times 3$ days) commenced **06.07.2021**.
 - **Complications:** neutropenic fever (cultures negative), grade 3 mucositis requiring TPN for 5 days, and a right forearm superficial thrombophlebitis at IV site (resolved). No cardiotoxicity.
 - **Response:** Day-28 marrow (03.08.2021) showed complete remission (**CR**), MRD negative by flow ($<0.1\%$).
- **Consolidation:** Intermediate-dose cytarabine (IDAC) $\times 3$ cycles from Sept–Nov 2021. Tolerated with expected cytopenias; required intermittent RBC/platelet support.
- **Transplant Decision:** Given t(9;11) and patient’s fitness, we recommended allogeneic hematopoietic stem cell transplant in first remission. His HLA-matched sibling was a suitable donor.
- **Allogeneic HSCT:** Performed **18.12.2021**. Conditioning: fludarabine/busulfan (myeloablative). GVHD prophylaxis: tacrolimus + methotrexate.
 - **Engraftment:** Neutrophils day +15, platelets day +20.
 - **Early complications:** grade I acute skin GVHD (palmar/plantar), managed with topical steroids; CMV PCR low-level viremia day +28, resolved with valganciclovir.
 - **Chimerism:** Day +30 mixed (92% donor myeloid), rose to 99% by day +100 without intervention.
 - **Disease status:** Marrow day +100: CR with MRD negative by KMT2A FISH and RT-PCR.

Current Status (follow-up through March 2024, ~32 months from induction start)

- Clinically well, ECOG performance status 0–1.

- **Laboratory values (05.03.2024):** WBC 5.1 (ANC 2.6), Hgb 13.4 g/dL, platelets 188 ×10⁹/L; creatinine 1.02 mg/dL; ALT 26, AST 21; ferritin 512 ng/mL (post-transfusion iron load).
- **MRD surveillance:** KMT2A rearrangement **undetectable** by sensitive assays on serial marrows (latest 12.2023) and peripheral blood PCR screens.
- **Immune reconstitution:** IgG 840 mg/dL; CD4 count improved to 420/μL.
- **GVHD:** Currently quiescent. We tapered tacrolimus off by Sept 2022. No chronic GVHD of mouth, eyes, or lungs.
- **Vaccinations:** Inactivated schedule restarted; received pneumococcal, Hib, tetanus, and annual influenza. COVID boosters current.

Comorbidities

- Hypertension (amlodipine 5 mg daily).
- Hyperlipidemia (rosuvastatin 10 mg nightly).
- Past smoker (20 pack-years; quit 2015); mild COPD with rare albuterol use.
- GERD (famotidine 20 mg at bedtime).

Medications

- Amlodipine, rosuvastatin, famotidine, cholecalciferol 1000 IU daily, PRN albuterol. No current immunosuppressants or antimicrobial prophylaxis.

Late Effects / Survivorship Considerations

- **Iron overload:** Ferritin trending down after four phlebotomies; we will continue therapeutic phlebotomy q6–8 weeks aiming ferritin <300 ng/mL.
- **Endocrine:** Screening TSH and fasting glucose normal.
- **Bone health:** Baseline DEXA was low-normal; calcium/vitamin D and weight-bearing exercise advised.
- **Cardiac:** Periodic echocardiography given anthracycline exposure; last LVEF 60% (08.2023).
- **Infection risk:** Routine prophylaxis no longer required; prompt evaluation for fever remains essential.

Plan Going Forward

- Hematology clinic follow-up every 3–4 months with CBC, CMP, and KMT2A MRD assay at least twice yearly for the next two years, then annually.
- Continue phlebotomy for iron reduction.
- Annual dermatology and ophthalmology given prior GVHD risk.
- Return sooner for any constitutional symptoms, bruising/bleeding, or recurrent infections.

In summary, Mr. Moore remains in **ongoing complete remission** with full donor chimerism after 1L induction/consolidation followed by allogeneic HSCT in first CR. There is no clinical or laboratory evidence of disease recurrence.

Please contact us if any issues arise.

Sincerely,
[Name], MD, PhD
Consultant Hematologist